

The Earnings Effect: Do Tech Stocks Actually Become More Volatile Around Earnings Time?

Data 101 · Rutgers University

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Interactive Website: sadhanavasanthakumar.github.io/data101-volatility

GitHub: github.com/SadhanaVasanthakumar/data101-volatility

The Question

Every quarter, companies like Meta, Apple, Amazon, Netflix, and Google report their earnings — how much money they made, how many users they have, what they expect going forward. These reports are called **earnings announcements**, and the days surrounding them are informally known as **earnings season**. Stock prices can swing wildly on these days: Netflix lost \$50B in market value overnight in April 2022; Meta fell 26% in a single session in February 2022.

We asked: is this just bad luck, or do earnings announcements *reliably* make these stocks more unpredictable, every quarter, year after year? And once a stock gets rattled by an earnings report, how long does the turbulence actually last?

The Data

We pulled daily stock prices for META, AAPL, AMZN, NFLX, and GOOGL from Yahoo Finance (January 2019 – December 2024, $\approx 1,508$ trading days per stock). We manually collected 24 quarterly earnings dates per company from Nasdaq’s public calendar (120 events total) rather than using automated databases, which are known to be unreliable for historical dates.

We measure each day’s movement as its **absolute daily return** — how much the stock went up or down, regardless of direction. A 3% day means the stock moved 3%, whether that was a gain or a loss. For each earnings date, we flagged a 5-day window (3 days before the announcement, the day itself, and 1 day after) as “earnings-adjacent” — information tends to leak before announcements, and markets often keep reacting the day after. Everything else was labeled “normal trading.”

What We Did — Course Topics Used

Data exploration & visualization. We started by just looking at the data: how much do these stocks typically move, what are their worst and best single days, how did their prices grow relative to each other over six years. Interactive charts for all of this are on the website.

Hypothesis testing. We used a **Welch’s t -test** to formally ask: is the average daily movement during earnings windows significantly higher than on normal days? Result — yes, for all five stocks, at a strict 1% significance level. Meta moves almost twice as much during earnings windows (3.2%) as on normal days (1.6%). The test rules out that this difference is random chance.

Modeling. We explored a **GARCH model** to understand how volatility behaves over time. The core insight it captures: a volatile day tends to be followed by another volatile day — calm periods and stormy periods both cluster together. By fitting this model to each stock, we found that after an earnings shock, *the turbulence doesn’t just disappear*. Netflix’s elevated volatility takes roughly 95 trading days (≈ 4.5 months) to decay to half its original size. Google recovers in about 9 days. We also used the model to predict future volatility and tested those predictions against what actually happened.

Bayesian inference. For Netflix (the most volatile stock), we ran a Bayesian version of the same model. Instead of giving one best-guess answer, Bayesian inference gives a *range* of plausible answers and shows how confident we should be in each. The result: forecast intervals that actually captured 93% of outcomes, compared to 86% for the standard model. Acknowledging uncertainty made the predictions more honest and more accurate.

Limitations

Our 5-day earnings window is a judgment call — different window sizes would shift the numbers slightly, though the core finding holds. We only studied five large US tech stocks, so results may not generalize to other sectors or smaller companies. The GARCH model also assumes returns behave somewhat like a bell curve, but in reality extreme days (like Netflix -43%) happen more often than any bell-curve model predicts. These are known trade-offs in financial modeling, not oversights.

How to Explore Our Work

Our main deliverable is the interactive blog:

sadhanavasanthakumar.github.io/data101-volatility

This is where the full project lives — it presents all of our findings as a public-facing blog post with charts you can interact with: rolling volatility over time, price growth across all five stocks, a correlation heatmap, hypothesis test results, and the modeled risk level for each company. Each section is written to be readable without a statistics background.

All code and data is on GitHub:

github.com/SadhanaVasanthakumar/data101-volatility

`analysis.r` is the full R pipeline that runs every analysis; `outputs/results.json` is the pre-computed output that powers the blog. To reproduce from scratch: install R with packages `quantmod`, `rugarch`, `jsonlite`, `xts`, `zoo`, `moments`, then run `Rscript analysis.r`.